

STATISTICAL QUALITY CONTROL

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Supplementary Reference: *Fundamentals of Quality Control and Improvement* by Amitava Mitra. John Wiley and Sons, 3rd Edition, 2008. ISBN: 978-0-470-22653-7.

Course Description and Objectives

The primary focus of this course is on statistical methods that can be used for controlling or managing the quality in manufacturing or service industry. Topics include Shewhart control charts for variables and attributes, process capability analysis, acceptance sampling plans, design of experiments, total quality management, and six sigma principles. Understanding, implementation, and interpretation of these concepts and methods will be covered.

Upon successful completion of this course, you will be able to

- Determine appropriate SQC tools and techniques to address various quality issues;
- Implement SQC tools and techniques in spreadsheet environments;
- Apply critical thinking and problem-solving skills in individual and team settings; and
- Understand how to manage and improve the quality of products or service.

Grade Evaluation

The grade you receive for the course is intended to certify your demonstrated proficiency in the course materials. Your grade will be based on a weighted evaluation of the following categories:

1. Class Participation	5%
2. In-Class Assignments	15%
3. Excel Assignments	10%
4. Quizzes	15%
5. Exams	50%
6. Mini Course Project	5%

OM375 COURSE SCHEDULE**Topics**

- 1 Course Introduction**
- 2 Module 1: Statistical Foundations**
 - Basic Statistics
 - Normal & Sampling Distribution
- 3 Module 2: Introduction to Quality Control**
 - DMAIC Process
 - Control Chart Basics
 - Phase I & Phase II Control Chart
- 4 Module 3: Other Control Chart Topics**
 - Trial vs Revised Control Charts
 - X-bar & s Charts
 - Proportion Conforming Output
 - X & MR Charts
 - p Chart
- 5 Module 4: Process Capability Analysis**
 - Process Capability
 - Hypothesis Testing & Process Improvement
- 6 Module 5: Acceptance Sampling Plans**
 - Acceptance Sampling Evaluation
 - Acceptance Sampling Construction
- 7 Module 6: Quality Improvement**
 - Regression Analysis
 - Design of Experiments
 - Six Sigma Principles
- 8 Course Project Q&A / Preparation**
 - Course Project Presentation**